Group 4 Network and Protocol Comments

Three network-related issues or extensions are considered below. Each comment identifies the observed behavior, the required processing change, and the relevant IRC protocol context.

1. Output backpressure and multi-line replies

A client that is actively reading can still be disconnected when one short JOIN request produces many JOIN, NAMES and terminator messages faster than its session writer drains them. The problem is a TCP application-layer queue burst, rather than a peer failure: a temporary full queue should apply backpressure while preserving message order. Request processing must wait for output headroom outside the shared IRC state lock, then enqueue each response without holding that lock. The session should retain a bounded timeout for a genuinely stalled reader, while PONG and QUIT remain responsive. This concerns IRC JOIN and NAMES processing and numerics 353/366; the 512-byte IRC line limit applies to each message, while TCP delivery may contain many messages.

2. Presence queries and complete bounded replies

An ISON request can fit within the 512-byte client-line limit even when the server's 303 reply becomes too long after adding its prefix, recipient nickname and CRLF. Returning only the names that fit would make an online user appear offline. The server therefore needs an all-or-error decision: measure the encoded complete reply and return one 417 ISON error when it cannot fit. The Bot must batch requests using both request and reply budgets, drain every response from a failed batch, and discard partial results on timeout or reconnect. The relevant IRC commands are ISON and numeric 303; 417 is used here as the implementation's explicit error for an oversized result.

3. Registration state, nickname reservation and liveness

A client can send NICK, reserve a nickname and then keep the connection alive with PING/PONG without ever completing USER registration. If registration has no independent deadline, the incomplete client can hold a nickname and session resources indefinitely. The server should timestamp connection creation, enforce a monotonic registration deadline that activity cannot extend, and remove the client and nickname atomically when it expires. Private routing and presence visibility should also require completed registration. This follows the IRC NICK/USER registration sequence and the PING/PONG liveness messages: a responsive TCP peer is not necessarily a registered IRC client.

References: RFC 2812, sections 3.2.2 (JOIN), 3.2.5 (PART), 3.2.6 (LIST), 3.4.2 (LUSERS), 3.7.2-3.7.3 (PING/PONG), 4.9 (ISON), and 5.1 (numeric replies).